Manasa Vijayaraghavan

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EDUCATION

University of Southern California, Los Angeles, CA — Expected May 2027
M.S. Applied Data Science

SRM Institute of Science and Technology, Chennai, India — Jun 2025
B.Tech. Computer Science Engineering

TECHNICAL SKILLS

Languages: Python, SQL
ML / Deep Learning: PyTorch, TensorFlow, XGBoost, scikit-learn, NLP, spaCy, LoRA, GRPO, SFT, differential privacy, federated learning, anomaly detection
LLMs / Agents: Anthropic Claude API, Claude Code, LangGraph, RAG, human-in-the-loop workflows
Data / Cloud: PySpark, Pandas, PostgreSQL, MySQL, MongoDB, MLflow, FastAPI, AWS, GCP (Vertex AI, Cloud Run), Azure AI, GitHub Actions, Git

EXPERIENCE

Data & Engineering Project — Stealth AI Startup *Oct 2026*
- Engineered **de-identification, pseudonymization, and PII pipelines** on **AWS** for a platform where company data is pseudonymized before any downstream use.
- Shipped **API connectors** from scratch for **OpenProject**, **n8n**, **Fathom**, and **Roundcube** and **Yahoo Mail** over **IMAP**; each is paginated, resumable, and tested end to end on real accounts.
- Traced a **Fivetran** status bug through logs and **AWS**, and fixed **Confluence** uploads, by replaying failures on a local database and adding automated tests.
- Built the company- and partner-side **feedback app** (web UI, **AWS** integration, **Sentry** and **Google Analytics**) on **Netlify**, **AWS Lambda**, **Postgres**, **S3**, and **GitLab**; ran a **gap analysis** reporting data as Delivered, Partial, or Missing.

AI Native Intern — Wipro *June 2026 – Aug 2026*
- Architected **agentic LLM workflows** for life insurance with **Python**, **LangGraph**, and the **Anthropic Claude API**: multi-step data gathering, validation, reasoning, and conditional routing across stateful graph nodes.
- Owned **Life Underwriting Intelligence** and **Life Fraud & Contestability Review** end to end, with **human-in-the-loop** approval gates before consequential decisions; integrated with **Guidewire**, **ServiceNow**, and **Majesco**; built with **Claude Code**.

Data Science Intern — Ford Motor Company *Feb 2025 – May 2025*
- Built a **multimodal emotion analysis system** over video, audio, and text (**DeepFace**, **Vosk**, **VADER**) on **GCP Vertex AI**, a **75%** efficiency gain.

Data Science Intern — iNextLabs *Aug 2024 – Oct 2024*
- Built NLP, anomaly detection, and **GenAI** solutions with **PyTorch** and **Azure AI** serving **500+** users, cutting issue resolution time by **40%**.

PROJECTS

When Does RL Beat SFT? GRPO vs. Fine-Tuning for Small LLMs *(CSCI 566 team project; Python, PyTorch, LoRA)*
- Compared **GRPO** and **SFT** on Qwen3.5-4B and Llama-3.2-3B at matched compute; wrote the shared **compute-budget** and **metrics** modules; Shortest Path training accuracy rose from about 10–30% to 80–91% under GRPO.
- Tracked **FLOPs**, **GPU-hours**, reward, accept rate, and truncation on **A100** GPUs so every run is compared fairly.

De-Identification & PII Pipeline *(Python, spaCy, Regex, Differential Privacy, Flower, PyTorch)*
- Detected names, emails, phone numbers, and financial identifiers with **spaCy NER** and custom **Regex**; masked, pseudonymized, or redacted them; applied **differential privacy** across 10+ distributed nodes with **Flower**.

Production MLOps Platform *(FastAPI, DistilBERT, MLflow, Docker, GCP Cloud Run, GitHub Actions)*
- Served a **DistilBERT** sentiment model behind **FastAPI**, with **MLflow** tracking, data drift detection, and **GitHub Actions** CI/CD to **GCP Cloud Run**.

Hotel Booking Cancellation Risk Prediction *(Python, scikit-learn; team of four, team leader)*
- Led a two-stage model on **119,390 bookings**: **K-Means** segments, then **Logistic Regression** and **Random Forest**.

FundedLabs *(Python, MySQL, MongoDB)*
- Solo app on **NIH and NSF grant data** to find funded labs, with **MySQL** and **MongoDB** components.

SignMate AI *(**Patent Filed** — Python, PyTorch, TensorFlow, OpenCV)*
- Assistive communication system for deaf and hard-of-hearing users; **patent filed**.